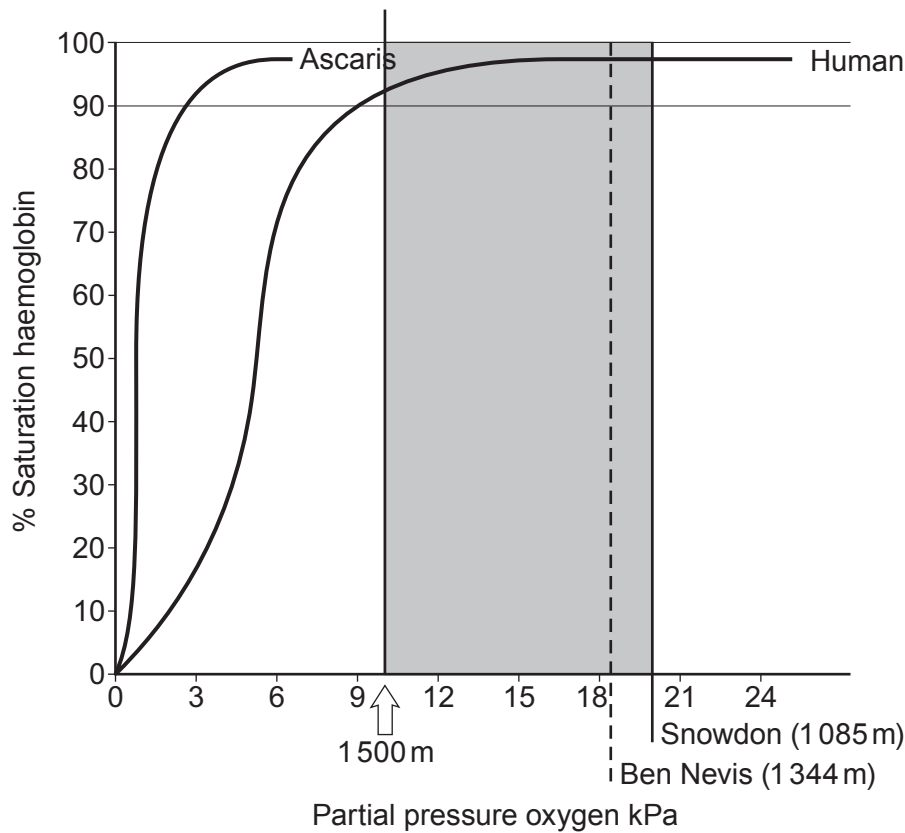


5. (a) The oxygen dissociation curves for human haemoglobin and for a parasitic worm (*Ascaris lumbricoides*), which lives in the gut of humans, are shown below.



People live at altitudes between sea level and approximately 5000m. Human health is not affected until the percentage haemoglobin saturation falls below 95%.

- (i) Explain the advantage to humans of the oxygen dissociation curve in the shaded region of the graph. [1]

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- (ii) Suggest how people become adapted to living at very high altitudes. [1]

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- (iii) Explain why the parasitic worm *Ascaris lumbricoides* has an oxygen dissociation curve as shown. [2]

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Examiner
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- (b) Crustaceans, such as crabs, have a molecule called haemocyanin in their blood which is used to transport oxygen from the respiratory surface to the respiring cells. It has similar properties to haemoglobin but contains copper instead of iron.
- (i) Haemoglobin and haemocyanin can be described as an example of convergent evolution. State what is meant by this term. [2]

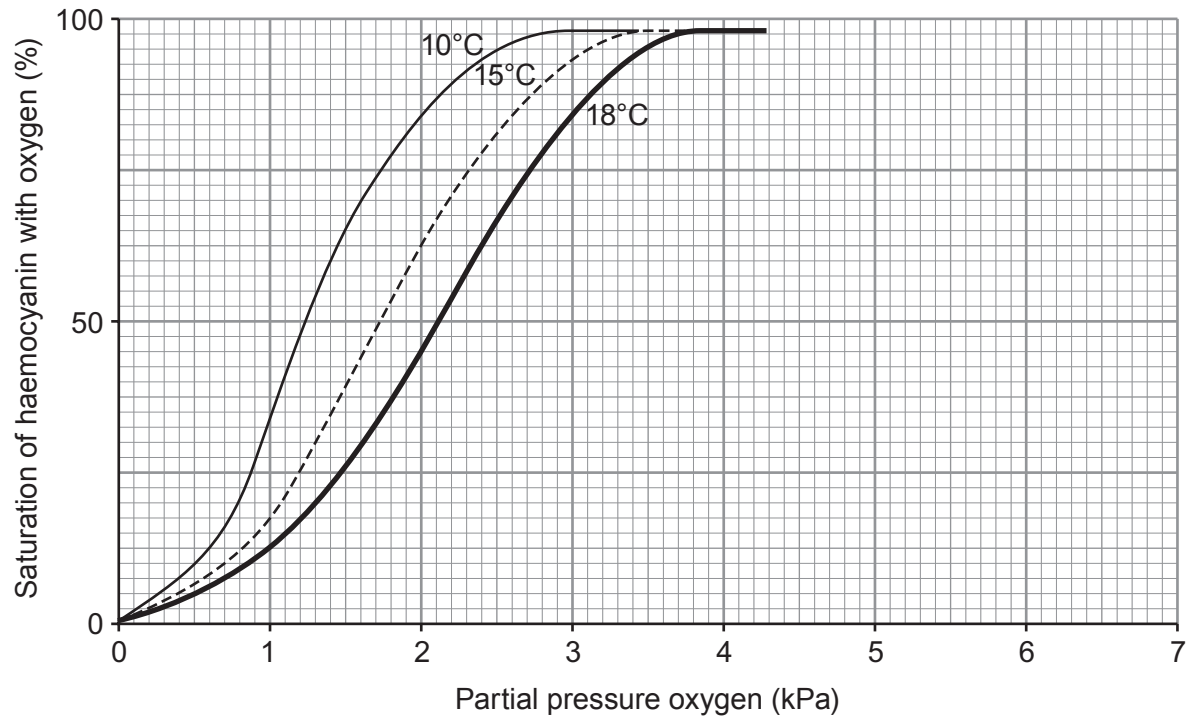
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The graph shows the effect of temperature on the dissociation of haemocyanin.



Crabs are often found in rock pools along the Welsh coast. In the Summer, water temperatures in rock pools can increase from 5 to 20°C during the day which reduces the oxygen content of the water. The body temperature of crabs varies with that of their surroundings.

(ii) What effect does an increase in temperature have on the partial pressure of oxygen at which the haemocyanin becomes 96% saturated? [1]

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(iii) The crab is living in water with a partial pressure of oxygen of 4 kPa and the partial pressure of oxygen in its tissues is 2.5 kPa. State an advantage to the crab when the temperature rises from 10°C to 18°C. [1]

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